

Secure-Document-Vault

This project is about a Secure Document Vault. The user accesses the web application via a website, log in or registers. After that they can upload or download files which will be encrypted and stored in the database.

Technologies

Frontend

- Lightweight client-side rendering without fancy animations.
- HTML, CSS, JS, BootStrap

Backend

- Async I/O ideal for file operations. Typescript adds extra type safety. Express is a node.js web framework
- Node.js with Typescript, Express.js

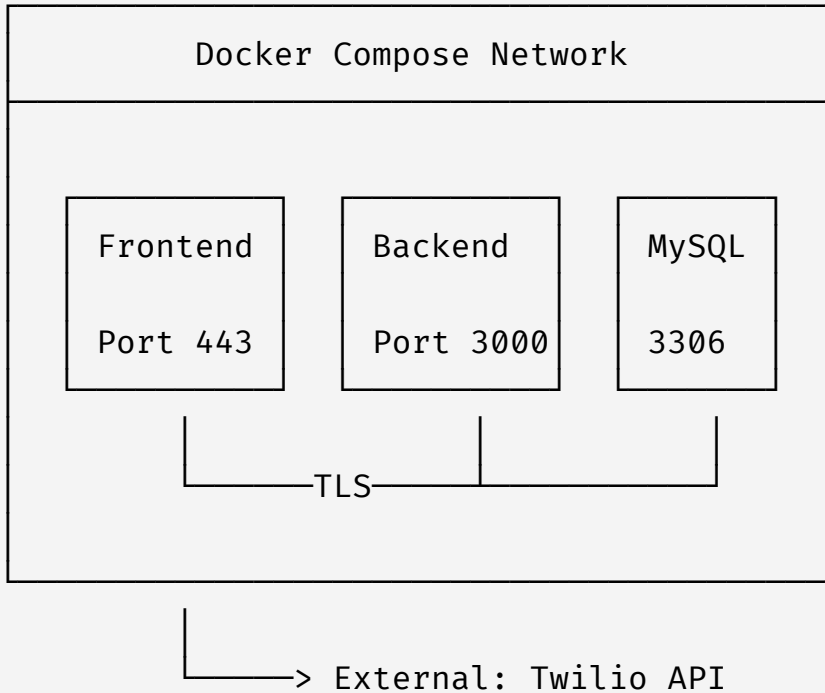
Third-Party

- Twilio for email, sms notification

Database

- MySQL 8

Container architecture



Cryptography

- TLS 1.3: encryption of HTTP traffic
- AES-256-GCM for file encryption, data at rest
- Argon2id library for hashing passwords
- JWT with HMAC-SHA256 for authenticated sessions (npm jsonwebtoken)
- npm crypto for general hashing and random key/IV generation
- *Post-Quantum*: ML-KEM for key encapsulation in future

Hosting

- Containerized with Docker Compose
- Frontend, Backend, DB separated
- Hosted possibly on DigitalOcean Droplet or self host

Domain

- Primary domain: vault.local
- Frontend: <https://vault.local:443>
- Backend api: <https://api.vault.local:3000>

Certificates

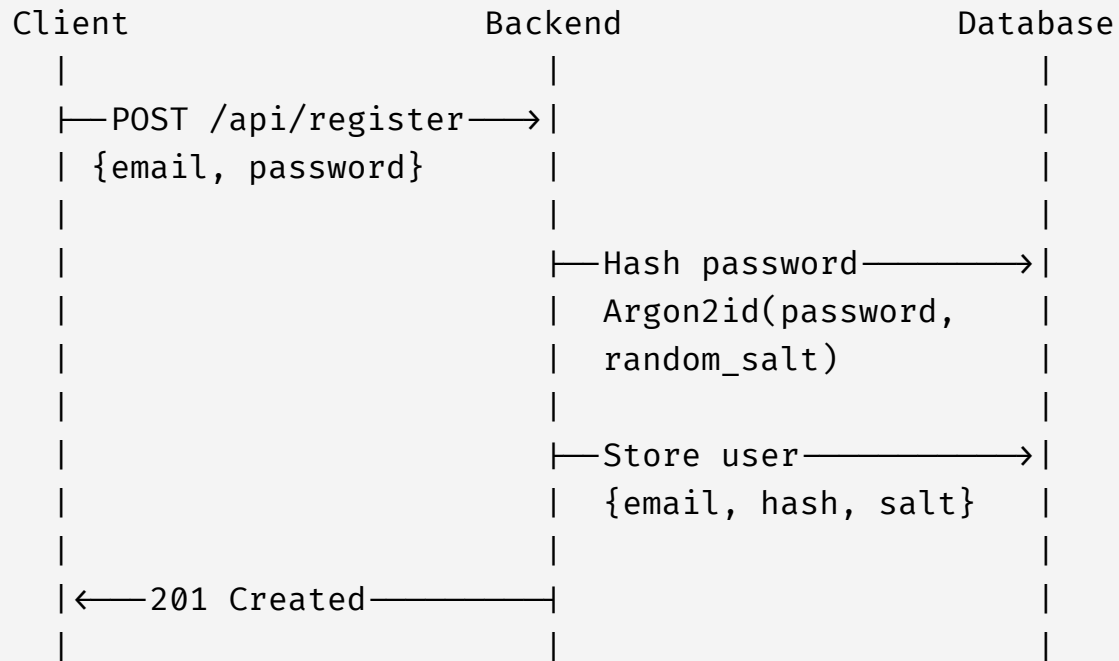
- Self-signed with OpenSSL

Functionalities

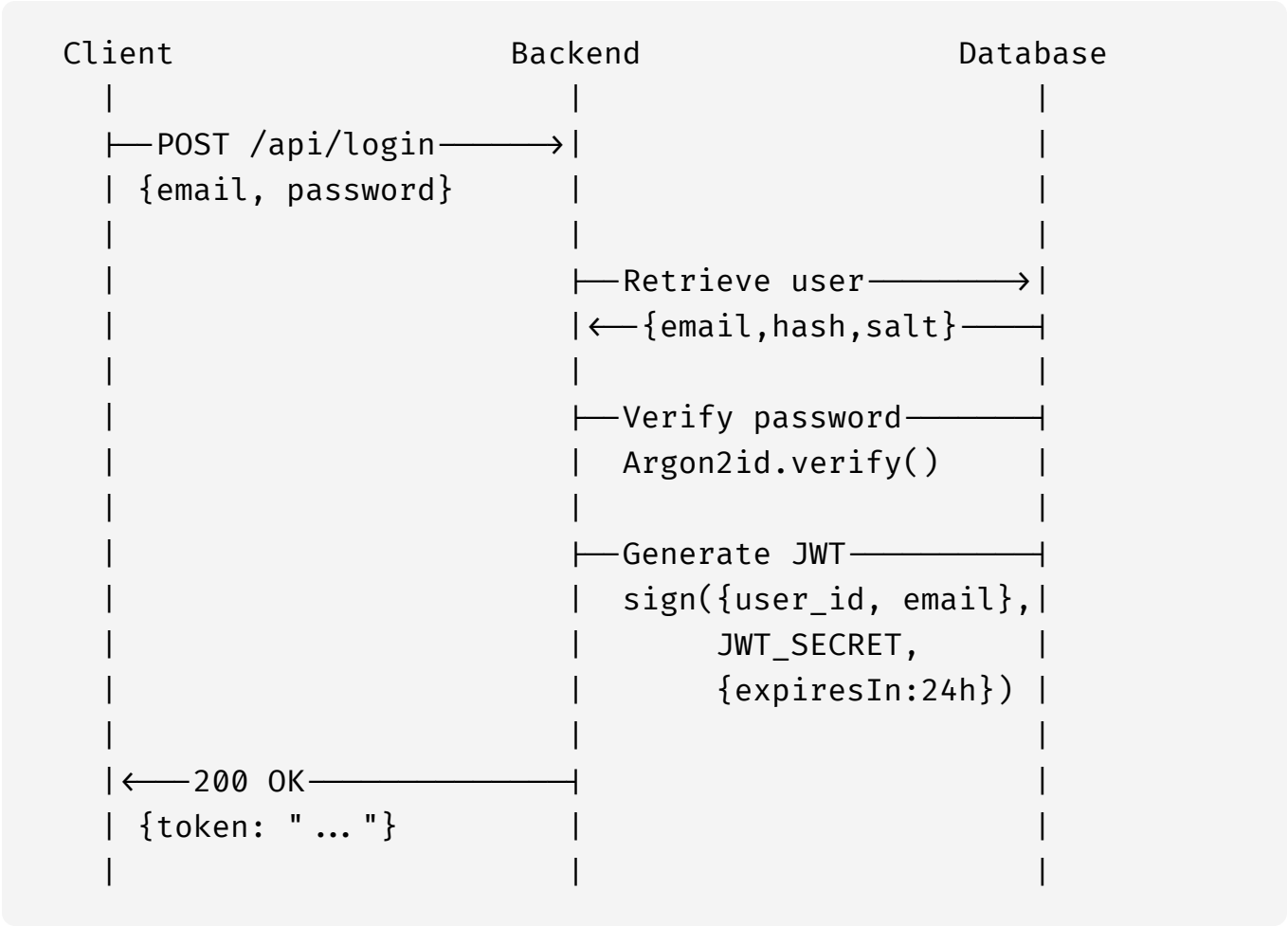
- Login, Authentication
- File upload
- File download

Data flows

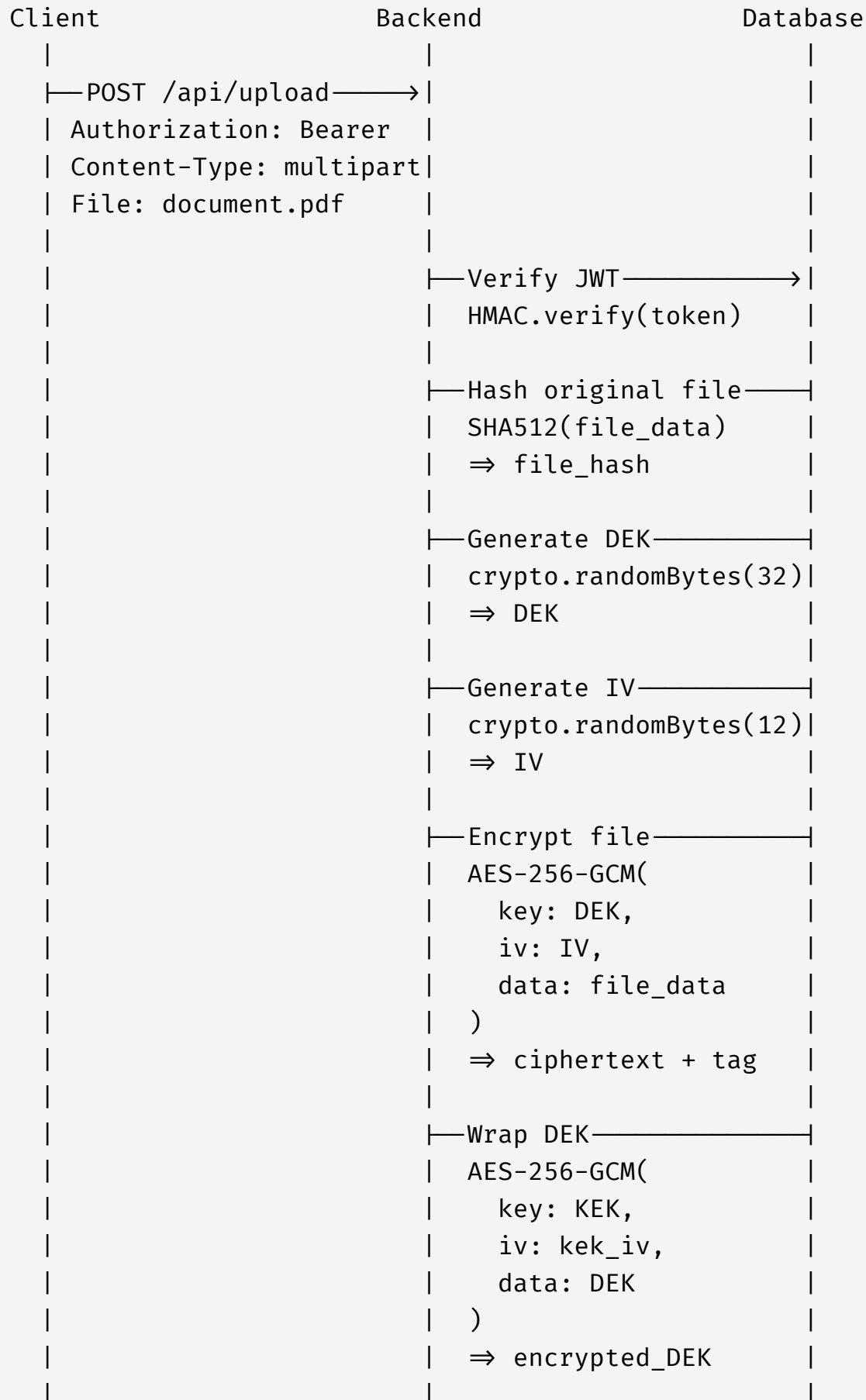
User registration



Login/Auth



File upload



```
|                                     |-----Store record----->|
|                                     | INSERT INTO files:         |
|                                     | {                           |
|                                     |   user_id,                 |
|                                     |   filename,                |
|                                     |   ciphertext,              |
|                                     |   encrypted_dek,          |
|                                     |   iv,                      |
|                                     |   auth_tag,                |
|                                     |   file_hash,               |
|                                     |   upload_timestamp         |
|                                     | }                           |
|                                     |<-----|
|                                     |
|                                     |-----Notify via Twilio----->|
|                                     | POST api.twilio.com        |
|                                     | "File uploaded"            |
|                                     |
|<-----200 OK-----|
| {file_id, message}      |
|
```

File download



{decrypted file data}	

File tree

Secure-Document-Vault

- ├── backend
 - ├── src
 - │ ├── models
 - │ ├── routes
 - │ ├── services
 - │ └── app.ts
 - ├── .env
 - ├── Dockerfile
 - ├── package-lock.json
 - └── package.json
- ├── frontend
 - ├── public
 - │ ├── css
 - │ └── js
 - ├── views
 - │ ├── index.html
 - │ ├── login.html
 - │ ├── logout.html
 - │ └── upload.html
 - ├── Dockerfile
 - ├── app.ts
 - ├── package-lock.json
 - └── package.json
- ├── .dockerignore
- ├── .gitignore
- ├── README.md
- └── cbom.json